MethylForge — epigenome (DNA-methylation) analysis (methylation calls → imprinting · episignature · DMR → clinical & research reports)

Short-read FASTQ
bisulfite / EM-seq
(paired-end)

Imprinted-DMR panel TSV
(bundled; locally validated)

Episignature reference
matrix + labels
(locally / ancestry-matched)

Genes BED
(DMR annotation)

Batch cohort QC dashboard
MultiQC-style HTML + summary TSV
[methylforge cohort-qc]

Stage 1 — Read QC & trimming
FastQC · fastp (bisulfite-aware)

Reference genome FASTA
(+ bisulfite spike-ins:
lambda · pUC19)

Stage 2 (short) — Methylation-aware alignment
Bismark (default) | bwa-meth · dedup

Illumina EPIC / 450K
IDAT (+ minfi sample sheet)

Long-read modBAM / FASTQ
Nanopore·PacBio native 5mC
(MM/ML tags; dorado upstream)

all samples

loci

reference

Stage 3 — Methylation extraction
MethylDackel (CpG + CHG/CHH) → bedGraph

Pre-called methylation
Bismark .cov · bedGraph ·
ENCODE bedMethyl · native TSV

Stage 0 — Array ingest (--mode array)
minfi/sesame: IDAT → beta + manifest
(bin/epic_process.R) → coord-keyed methyl
[methylforge array]

Stage 2 (long) — Align + pileup
minimap2 -y → modkit pileup → bedMethyl

Per-CpG methylation
(normalised schema: chrom·pos·meth·cov·beta·context)

cohort matrix

Stage 4 — Methylation QC
conversion (spike-in / CHH) · coverage ·
global meth by context · inferred sex · gates
[methylforge qc]

Stage 5 — Imprinted-DMR analysis
loss/gain vs per-locus normal → disorder
+ MLID flag
[methylforge imprinting]

Stage 7 — Episignature classification
nearest-centroid (default) | sklearn SVM
calibrated confidence + inconclusive zone
[methylforge episign · build-reference]

Stage 6 — Differential methylation (cohort)
DMP: Mann-Whitney/Welch-t on M-values + BH-FDR
→ merge → DMR
[methylforge cohort → dmr]

Stage 8 — Region annotation
nearest gene / context (genes BED)

Stage 9 — Report engine
clinical (GBCF/Noguchi, ACMG colours) +
research (full DMR detail) · results.json
[methylforge run]

Stage 9b — Excel findings workbook
Findings · Imprinting loci · QC
[methylforge findings-xlsx]

FamilyForge export
candidates.json-compatible + epi_findings TSV
(tiered A/B/C)

Stage groups

- Array ingest / cohort QC
- Read QC / trimming
- Alignment
- Extraction / methylation calls
- Methylation QC
- Imprinting / episignature
- DMR / annotation
- Reporting
- FamilyForge hand-off
- External input / resource (dashed = drives/labels)

[methylforge <cmd>] = runnable standalone
(methylforge run = single-sample engine)